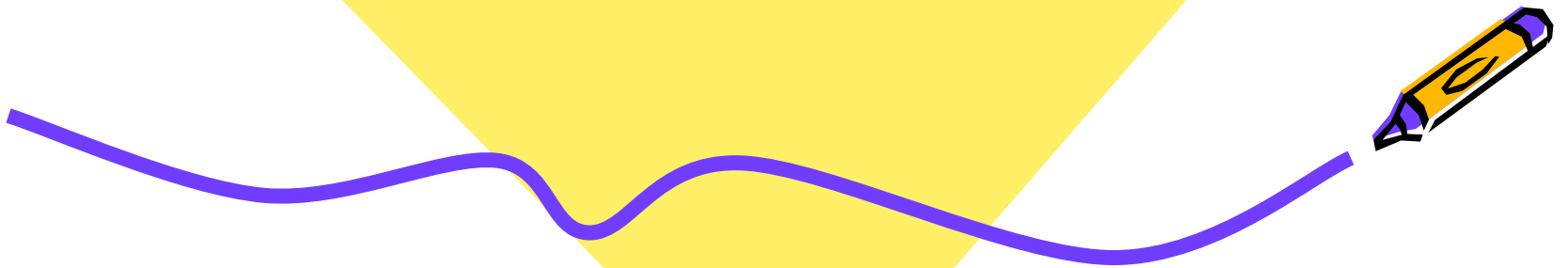




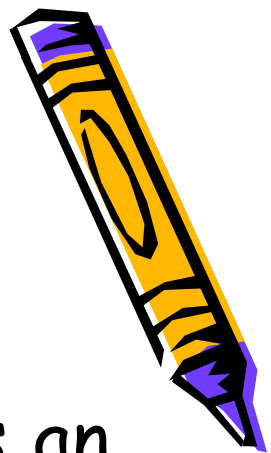
Projectile Motion



What is a Projectile?

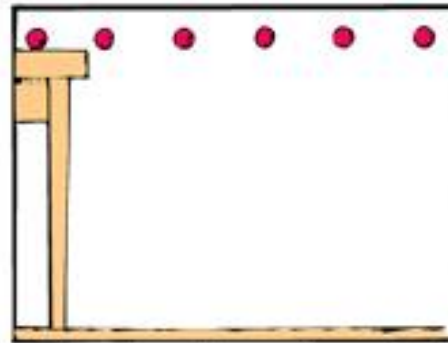
If we ignore air resistance, a projectile is an object upon which the only force acting is gravity.

This could mean dropping a rock, throwing a ball straight up or kicking a football. The rock, ball and football are all projectiles.

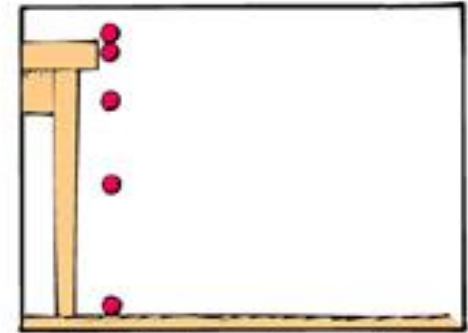


What is Projectile Motion?

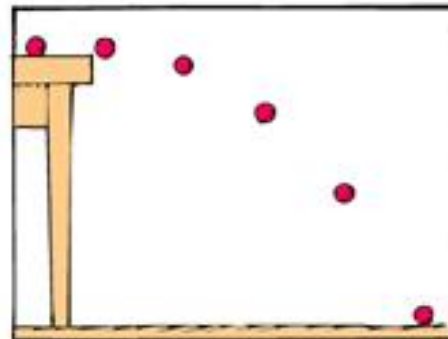
Projectile motion is a combination of uniform horizontal and freefall vertical motion



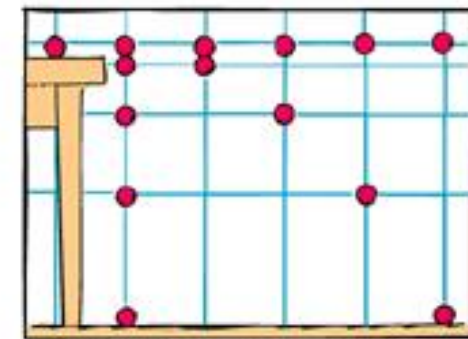
Horizontal motion with no gravity



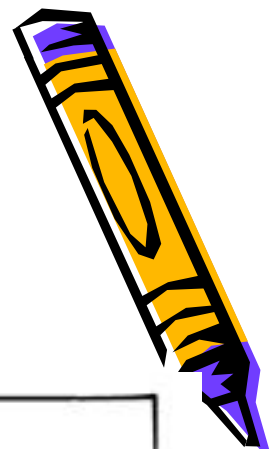
Vertical motion only with gravity



Combined horizontal and vertical motion

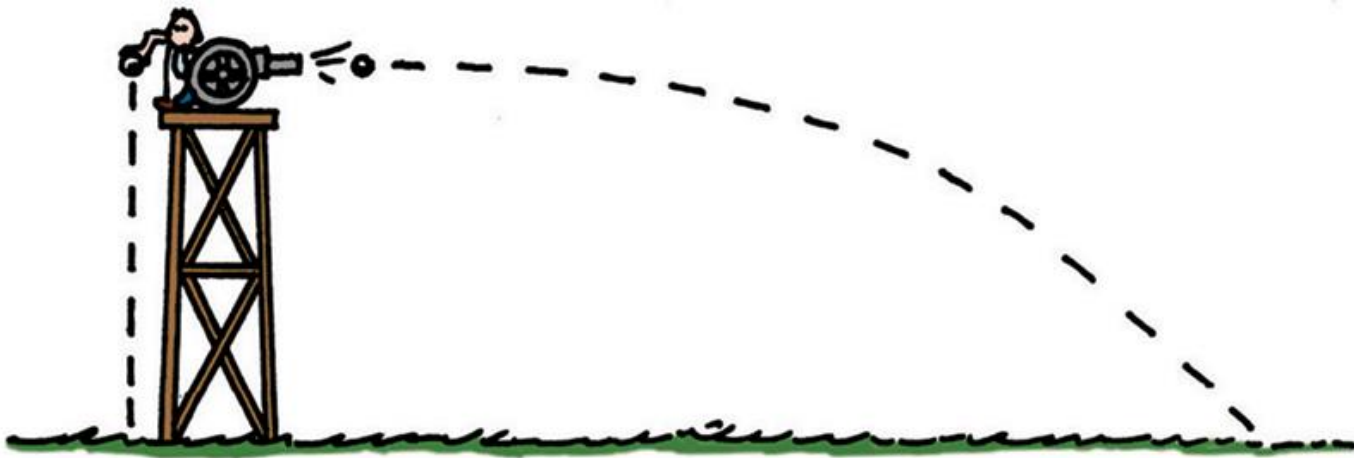


Superposition of the preceding cases



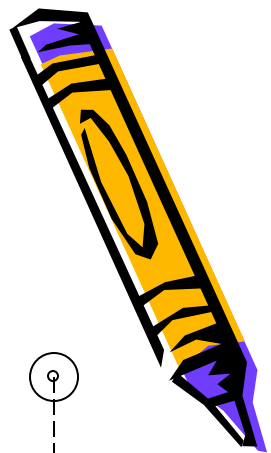
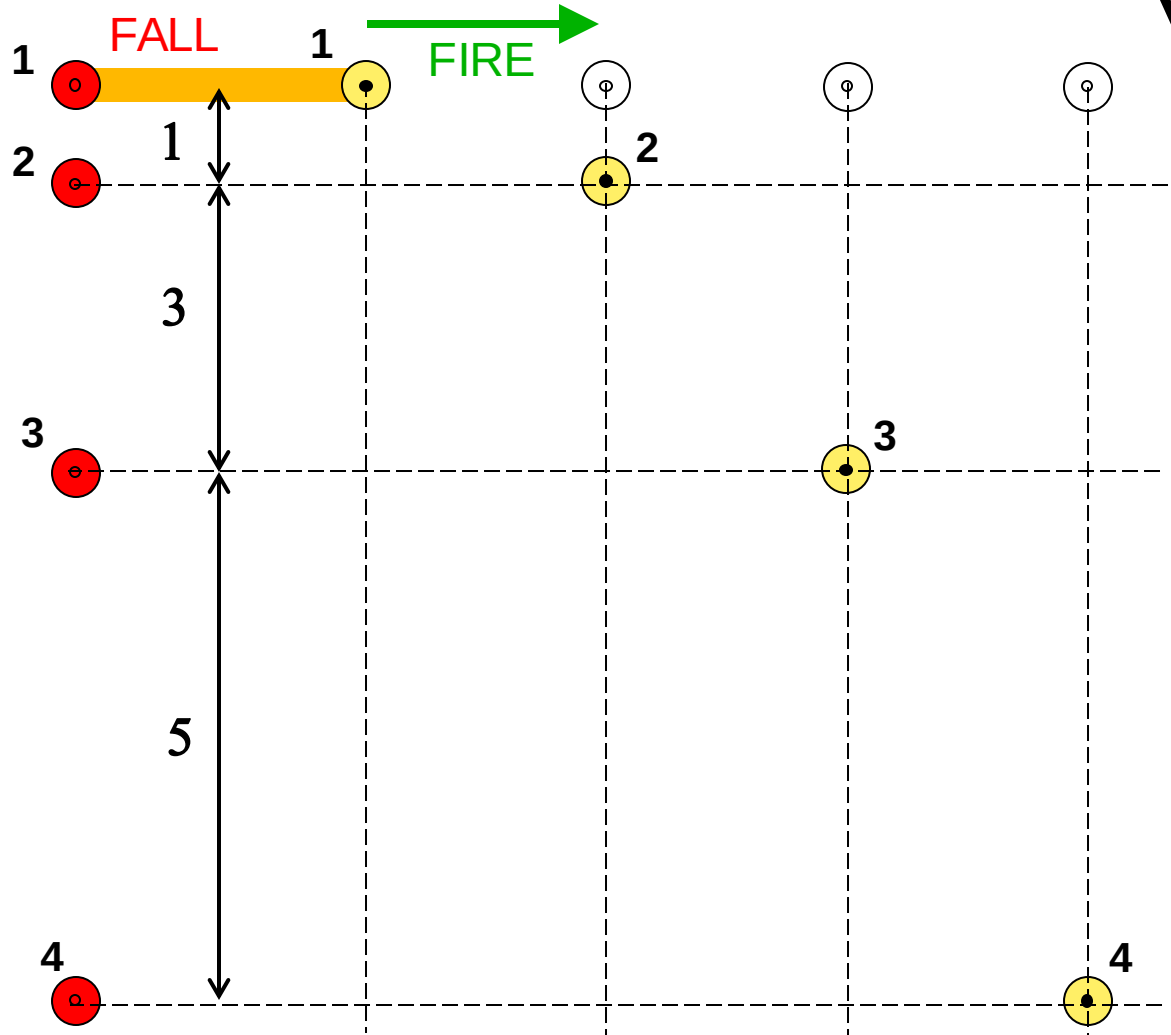
Test Your Thinking

At the instant a cannon fires a cannonball horizontally over a level range, another cannonball held at the side of the cannon is released and drops to the ground. Which strikes the ground first?



Cannon balls

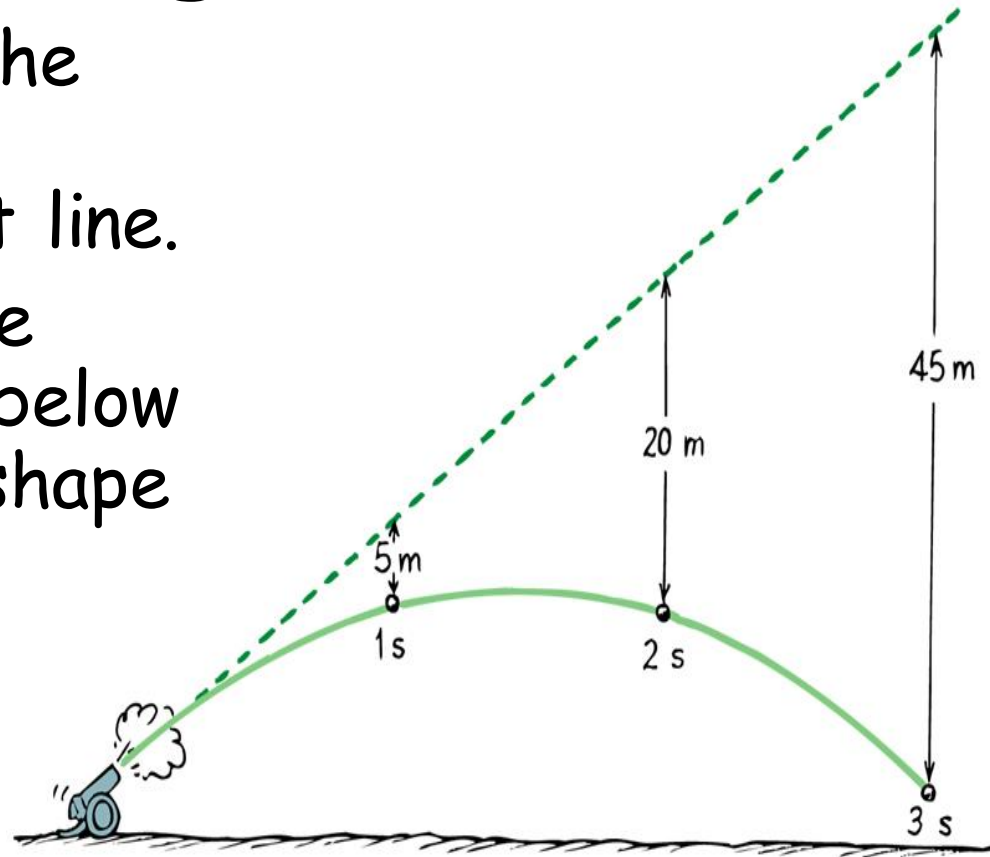
One ball is released and falls straight down.
Other ball is fired horizontally.
At all times the balls are at the same height.
They hit the ground at the same time.



Projectiles launched at an angle

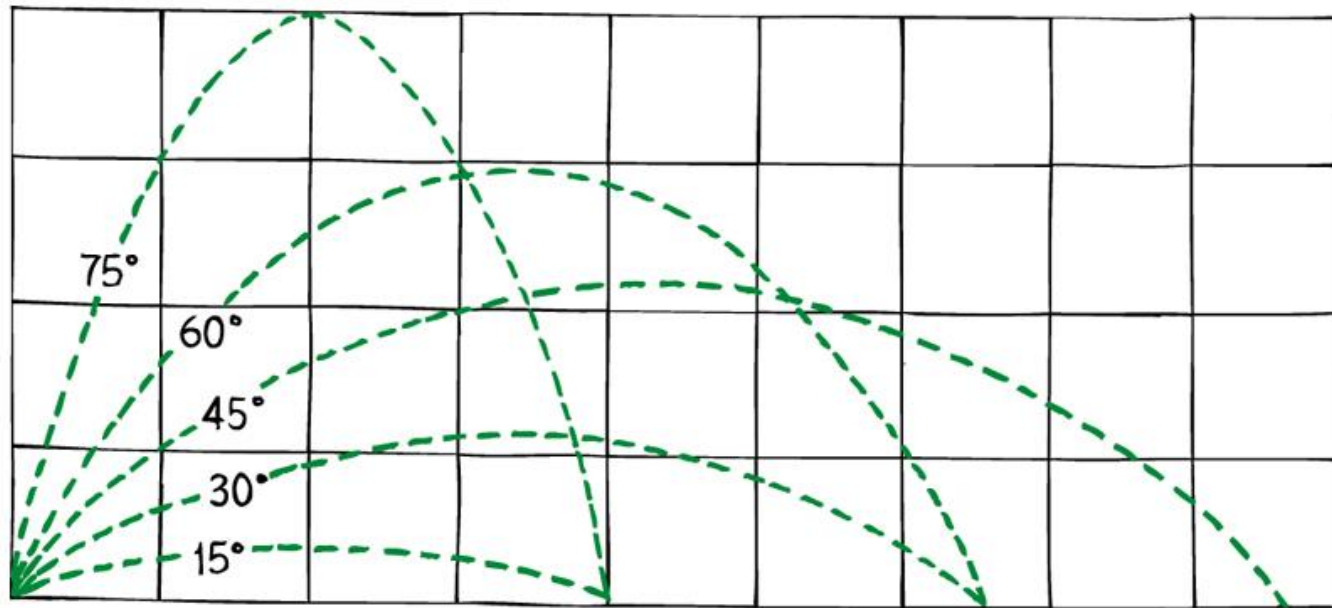
With no gravity, the projectile would follow a straight line.

Due to gravity, the projectile falls below this line in the shape of a parabola.



Maximum Range

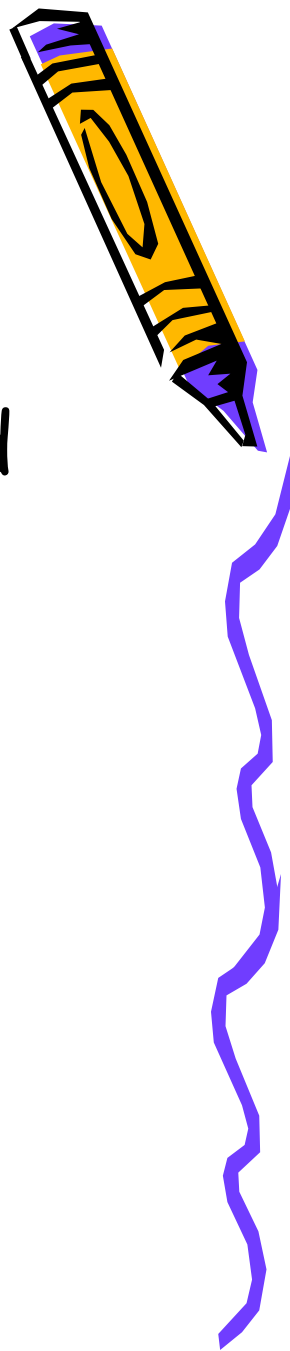
- The maximum range of a projectile is at a 45° angle when no air resistance is present.



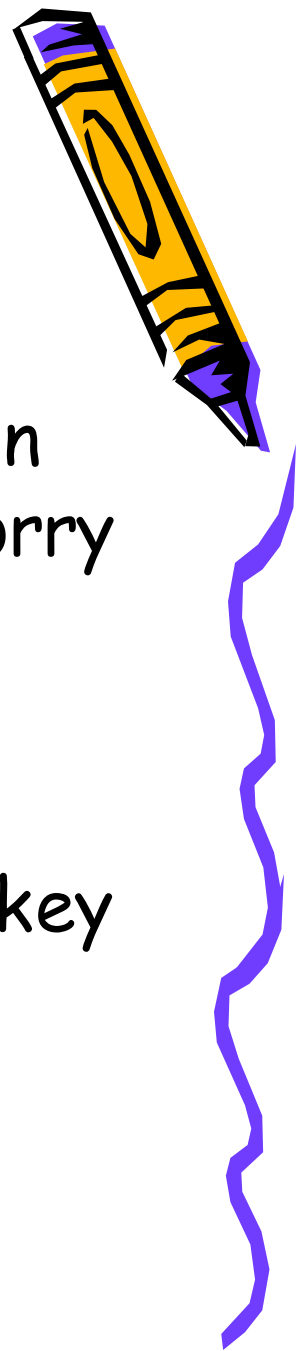
Air Resistance

If a projectile is effected by a head wind, the horizontal range will decrease.

If a projectile is effected by a tail wind the horizontal range will increase.



Projectile Motion Questions



The first step in tackling projectile motion questions is to draw a picture. Don't worry about the calculations yet... just draw a detailed, labelled picture of the information given.

Then transfer this information to a data key
- now you are ready to tackle the calculation.



Projectile Motion Calculations



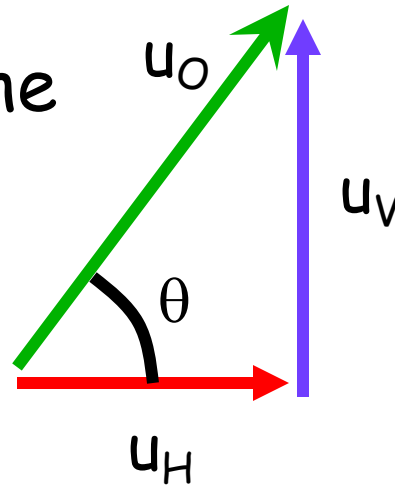
The first step to completing projectile motion calculations is to remember that the motion is occurring in two components, horizontally and vertically.

As a result these two components must be calculated separately.

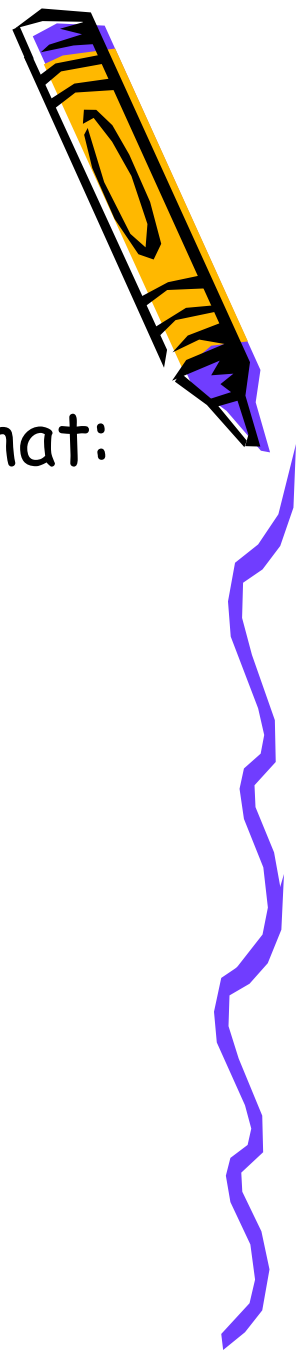


Initial Velocity

The initial velocity, u_0 of the projectile is the resultant vector of the horizontal and vertical components of the initial velocity.



Initial Horizontal and Vertical Components of Velocity



By using trigonometry you can work out that:
The initial horizontal velocity u_H is:

$$u_H = u_0 \cos \theta$$

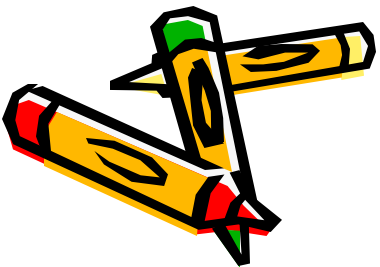
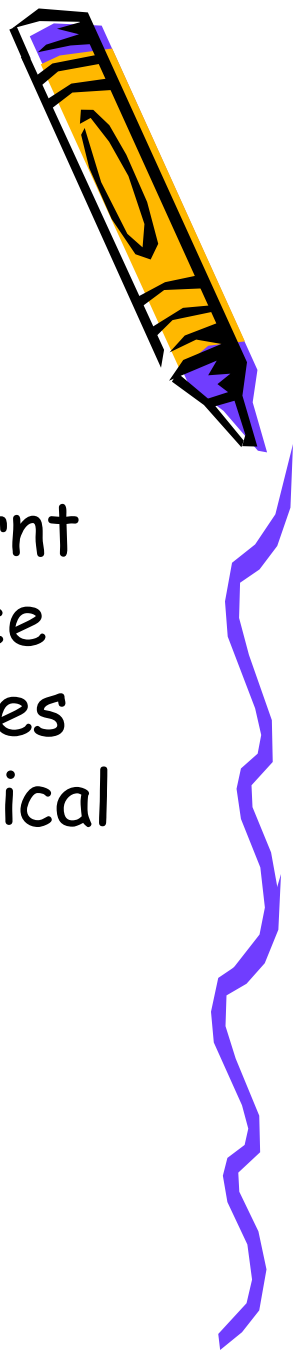
The initial vertical velocity u_V is:

$$u_V = u_0 \sin \theta$$

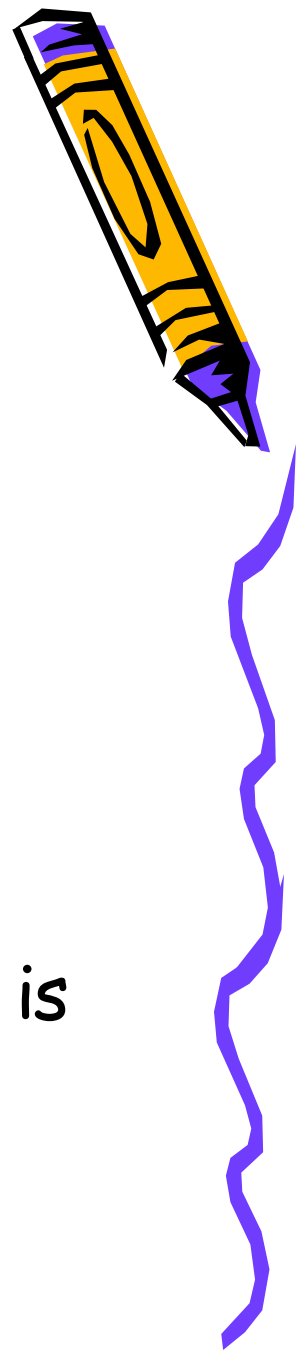


What's next?

All projectile motion calculations use the exact same movement formulas you learnt in Year 11 Physics. The major difference is that these formulas can only use values from one component i.e. either the vertical component or the horizontal component NOT both!



Motion formulas revisited



There are three main motion formulas:

$$v^2 = u^2 + 2as$$

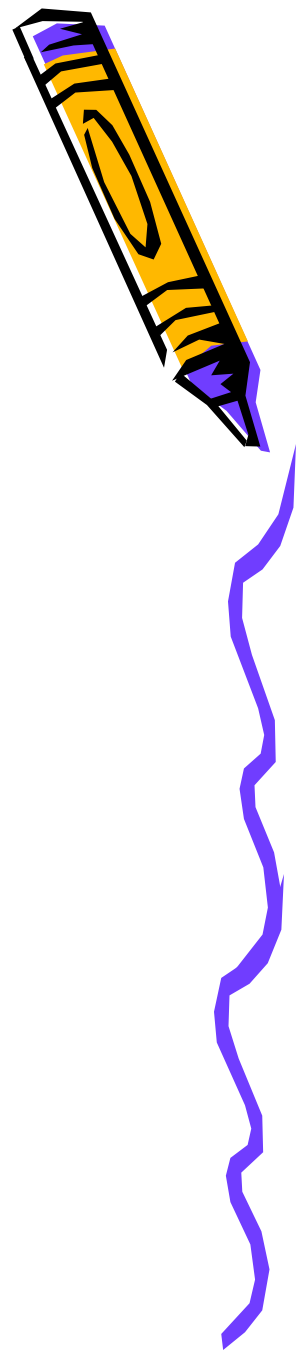
$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

a is the acceleration due to gravity which is 9.80ms^{-2}



Vertical component formulas



When using the formulas in the vertical component, assume that all downward values are negative. This makes the formulas become:

$$v_v^2 = u_v^2 - 19.6s_v$$

$$v_v = u_v - 9.8t$$

$$s_v = u_v t - 4.9t^2$$



Horizontal Component Formulas



There is no horizontal acceleration. As a result the Initial horizontal velocity is equal to the Final horizontal velocity.

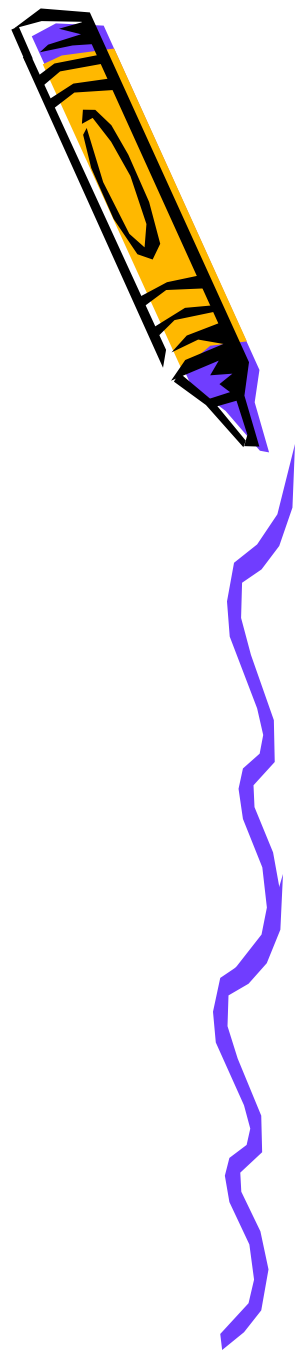
This means there is really only one formula we can use for the horizontal component:

$$s_H = u_H t$$

The other part of the formula has been removed as it would return a value of zero.



Things to remember



Time is the same in the horizontal and vertical components

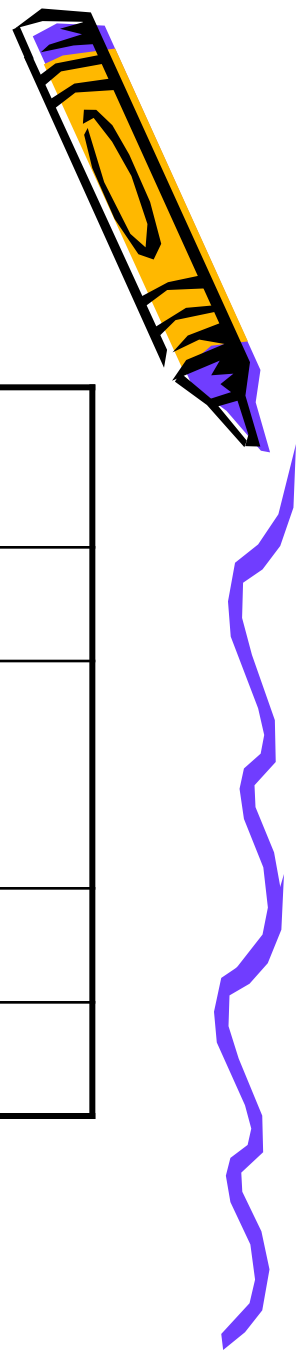
At the top of the parabola, the vertical velocity is zero.

You can not calculate the vertical or horizontal displacement using trigonometry. You must use the motion equations to calculate this.



A helpful table

Variable	Vertical Component	Horizontal Component
Acceleration (a)	-9.80ms^{-2}	0ms^{-2}
Initial Velocity (u)		
Final velocity (v)		
Displacement (s)		
Time (t)		



A Very Easy Question

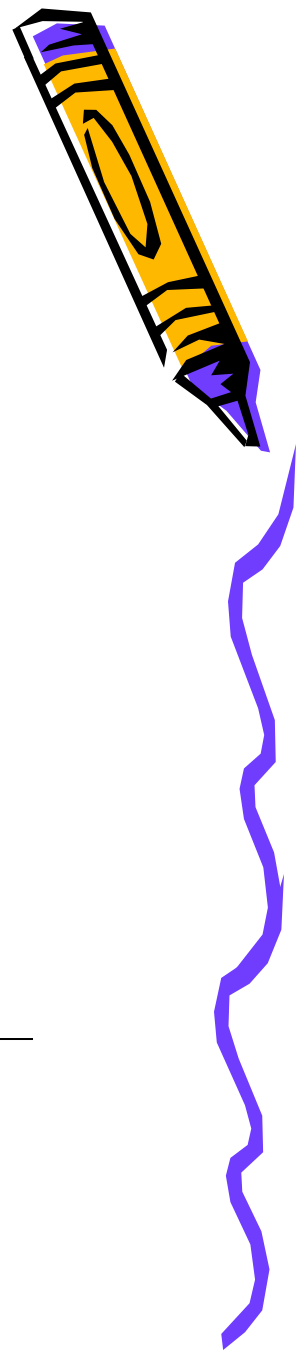
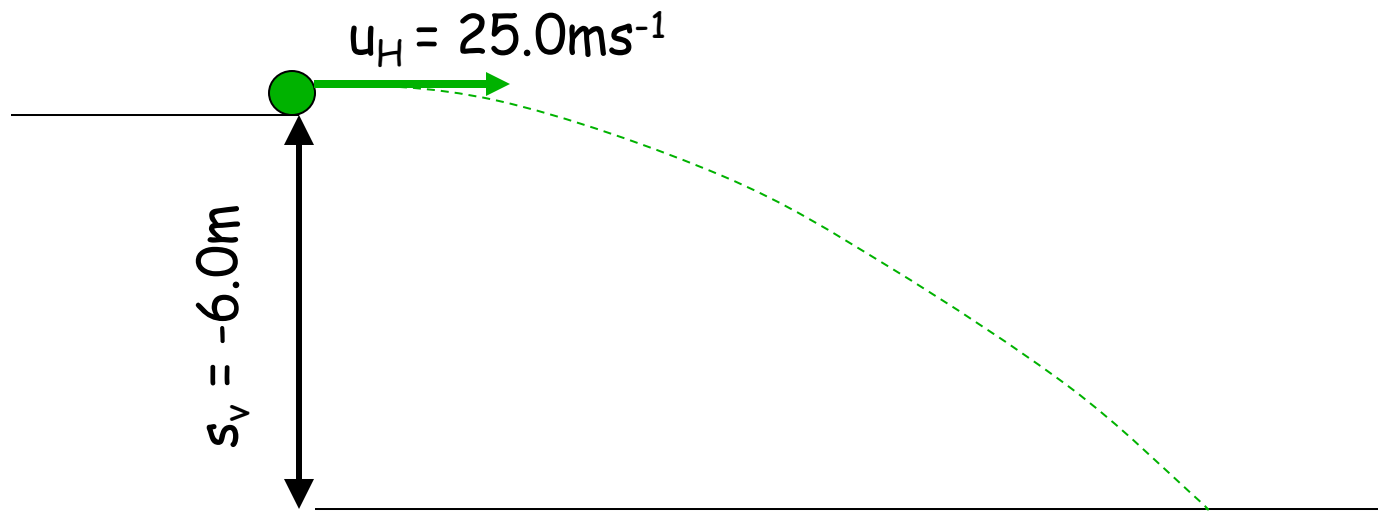


Brian was practising golf at a driving range. He was on an elevated tee 6.0m above the fairway and struck a ball so that it left his club with an initial horizontal velocity of 25.0ms^{-1} .

- a) How long after the ball leaves the club, will it land on the fairway?
- b) What horizontal distance will the ball travel before striking the fairway?

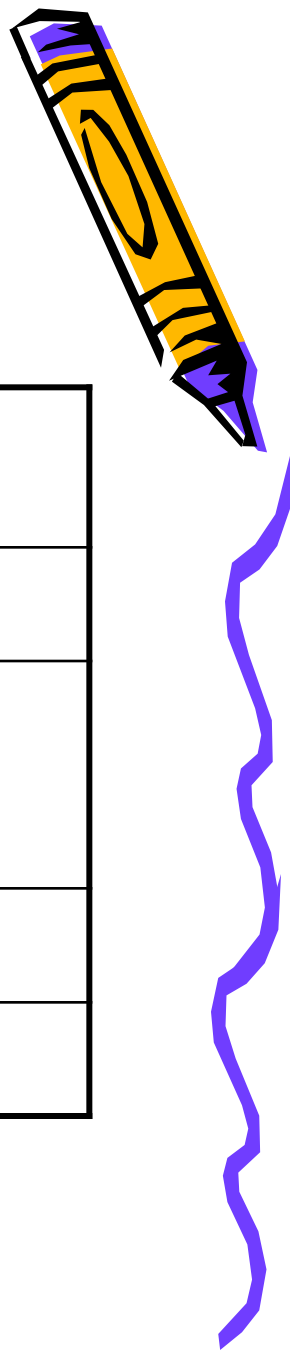


Step 1 - Draw a picture



Step 2 - Data key

Variable	Vertical Component	Horizontal Component
Acceleration (a)	-9.80ms^{-2}	0ms^{-2}
Initial Velocity (u)	0ms^{-1}	25.0ms^{-1}
Final velocity (v)		
Displacement (s)	-6.0m	
Time (t)		



Use the data to answer the questions

a) Use the formula:

$$s_v = u_v t - 4.9 t^2$$

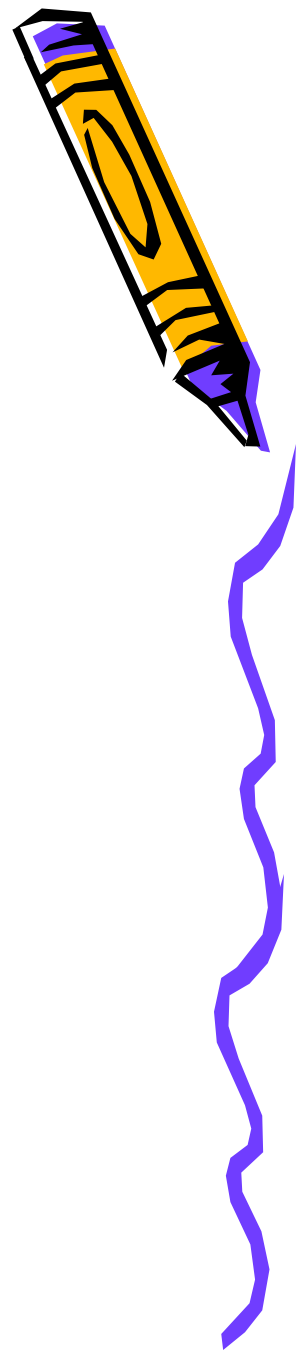
$$-6.0 = 0.t - 4.9 t^2$$

$$-6.0 = -4.9 t^2$$

$$-6.0 / -4.9 = t^2$$

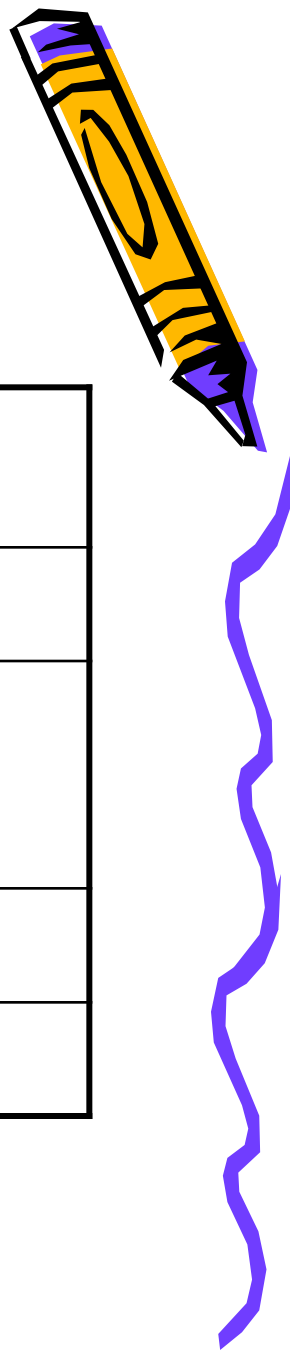
$$t^2 = 1.22$$

$$t = 1.11s$$

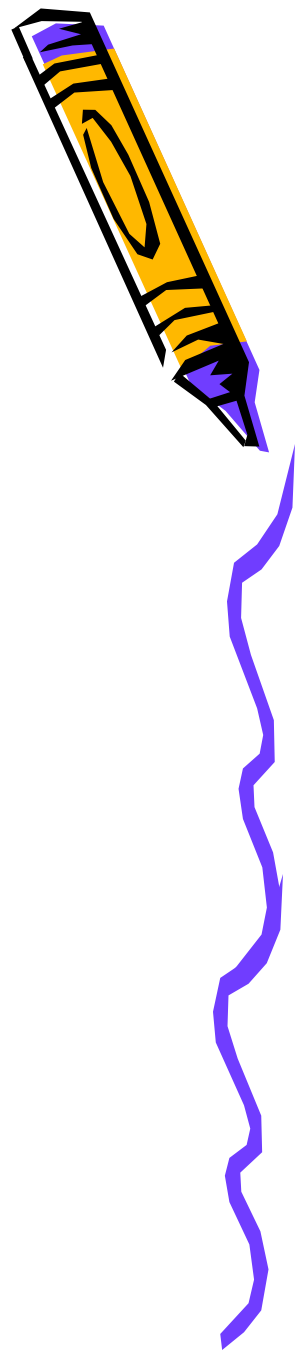


Step 2 - Data key

Variable	Vertical Component	Horizontal Component
Acceleration (a)	-9.80ms^{-2}	0ms^{-2}
Initial Velocity (u)	0ms^{-1}	25.0ms^{-1}
Final velocity (v)		
Displacement (s)	-6.0m	
Time (t)	1.11s	



Use the data to answer the questions



b) Use the formula:

$$s_H = u_H t$$

$$s_H = 25.0 \times 1.11$$

$$s_H = 27.7\text{m}$$



A little more complicated question



Blake kicked a soccer ball off the ground at an angle of 30 degrees at 22.0ms^{-1} .

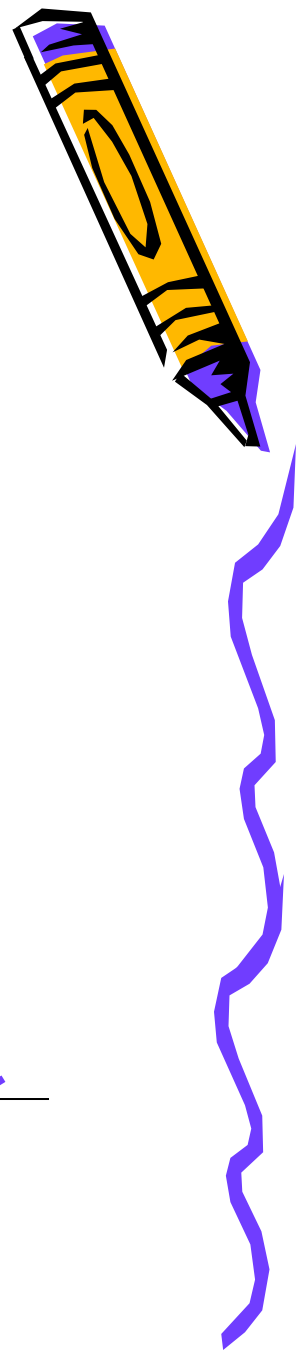
- a) What is the maximum height the ball will reach?
- b) Calculate the horizontal range of the soccer ball kick.



Step 1 - Draw a picture

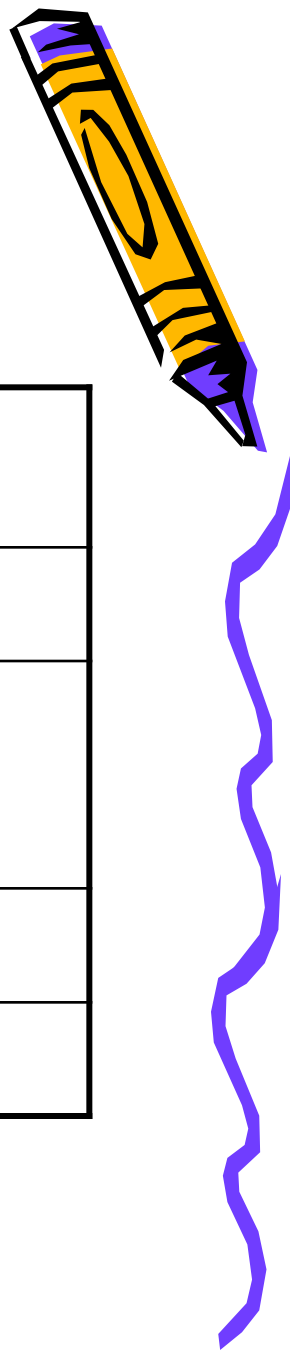
$$u_0 = 22.0 \text{ ms}^{-1}$$

$$\theta = 30^\circ$$

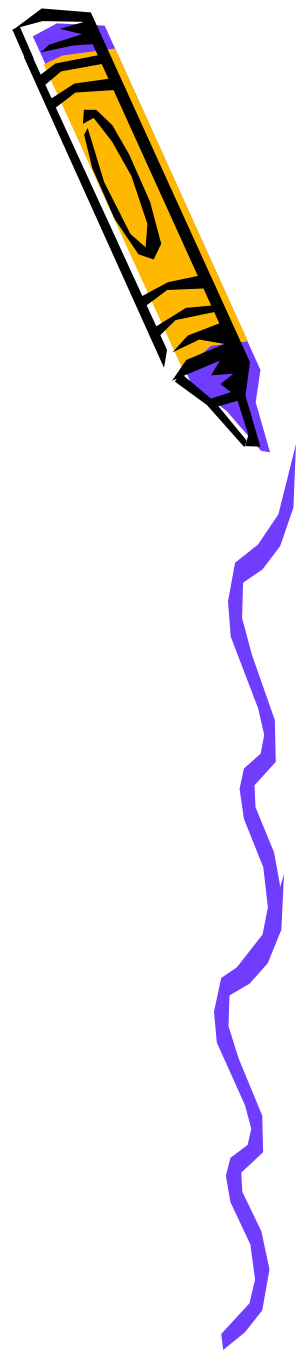


Step 2 - Data key

Variable	Vertical Component	Horizontal Component
Acceleration (a)	-9.80ms^{-2}	0ms^{-2}
Initial Velocity (u)		
Final velocity (v)		
Displacement (s)		
Time (t)		



Step 3 - Calculate initial component velocities



The initial horizontal velocity u_H is:

$$\begin{aligned}u_H &= u_0 \cos \theta \\&= 22.0 \cos 30 \\&= 19.1 \text{ms}^{-1}\end{aligned}$$

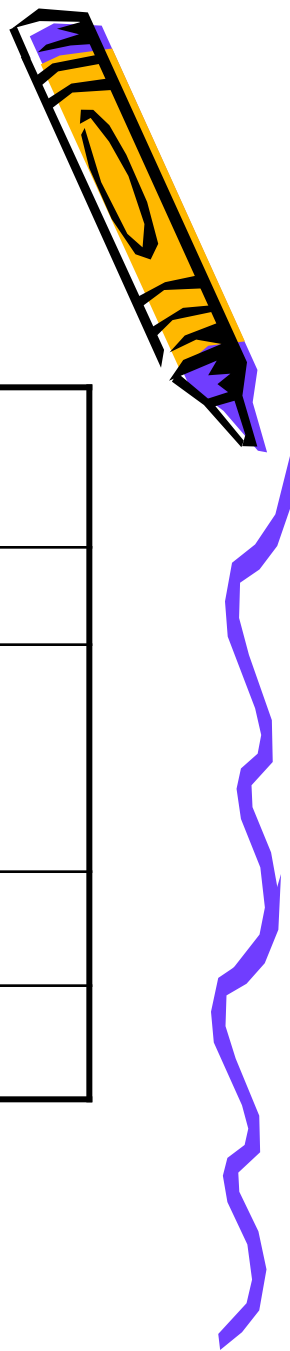
The initial vertical velocity u_V is:

$$\begin{aligned}u_V &= u_0 \sin \theta \\&= 22.0 \sin 30 \\&= 11 \text{ms}^{-1}\end{aligned}$$



Step 2 - Data key

Variable	Vertical Component	Horizontal Component
Acceleration (a)	-9.80ms^{-2}	0ms^{-2}
Initial Velocity (u)	11.0ms^{-1}	19.1ms^{-1}
Final velocity (v)		
Displacement (s)		
Time (t)		



Step 4 - Use the data to answer the questions



- a) At maximum height, the vertical velocity is zero, so use the equation:

$$v_v^2 = u_v^2 - 19.6s_v$$

$$0 = 11^2 - 19.6s_v$$

$$19.6s_v = 121$$

$$s_v = 121/19.6$$

$$= 6.17\text{m}$$



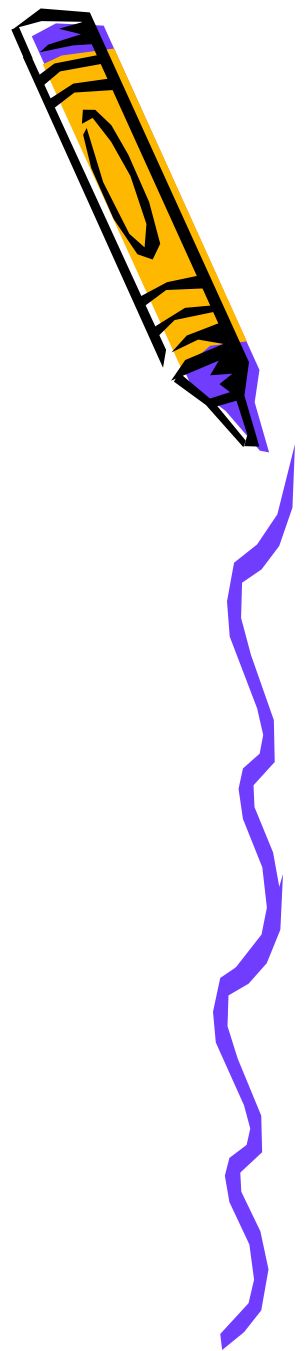
Step 4 - Use the data to answer the questions



- b) To find the range of the soccer ball we need to know how long its trajectory lasted. We can calculate this by finding out how long it would take for the ball to reach the top of its trajectory and then multiplying the answer by two. This can be done because this particular parabola is symmetrical.



Step 4 - Use the data to answer the questions



b) Use the equation:

$$v_v = u_v - 9.8t$$

$$0 = 11 - 9.8t$$

$$9.8t = 11$$

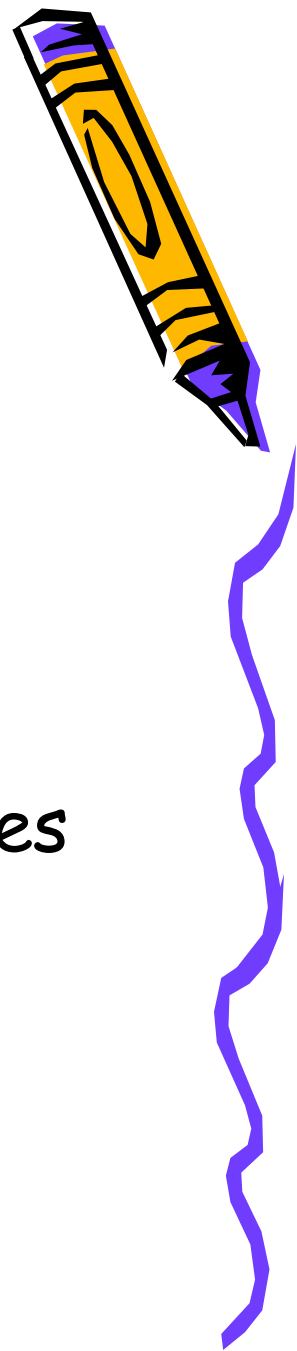
$$t = 11/9.8$$

$$t = 1.12s \text{ for half parabola}$$

$$t = 2.24s \text{ for entire range}$$



Step 4 - Use the data to answer the questions



b) Now we can use the formula:

$$s_H = u_H t$$

$$s_H = 19.1 \times 2.24$$

$$s_H = 42.9\text{m}$$

* Make sure you use exact calculator values in your calculations.. not your rounded values.

